Selective Language Damage with Parameter-Decomposition Subcomponents

Sergei Solovev

Independent researcher

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I study selective language damage from editing individual adVersarial Parameter Decomposition (VPD) subcomponents in a four-layer, 67M-parameter model, measuring the effects on a target language and its close relative. A fixed rule uses target and English text to select one subcomponent per target. Six edits across three pairs (German and Dutch, Spanish and Portuguese, French and Italian) raise target cross-entropy (CE) by 0.5 or 1 nat/token on validation text. I compare them with rank-one LoRA, full-matrix fine-tuning and neuron edits calibrated the same way.

On 58 European Parliament session dates evaluated once after the edits were fixed, the subcomponent edits raised the related language's CE by at most 0.0053 nat/token. All 18 simultaneous 95% intervals for the differences from the comparators lay below zero. The subcomponent edits therefore did less related-language damage at matched validation effects. A post hoc check found achieved target effects to agree within ±0.1 nat/token in 15 of 18 comparisons. The learned updates preserved English output better.

Exploratory comparisons on previously evaluated development dates generally narrowed the gaps to learned updates trained for 2,000 steps, while retaining the subcomponent edits' advantage at the tested budgets. The subcomponent edits also did less related-language damage than a whitened rank-one selector. The result concerns selective fixed edits in this model; the dictionary's contribution, the mechanism and generalization of the selection rule remain unresolved.

1 Introduction

A model edit that changes one behaviour usually changes others too. Model-editing work evaluates *locality*: inputs outside an edit's scope should keep their predictions (Yao et al. 2023, sec. 2). ROME, for example, scores an edited fact by whether the model retains facts about nearby subjects (Meng et al. 2022, sec. 3.3).

I use deliberate language damage as a controlled test of edit selectivity: each edit is calibrated to the same target-language damage on validation text, then evaluated for its effects on another language. A demanding neighbour for a language edit is a closely related language. Zeroing the parameters or neurons most important for one language also degrades its relatives: Spanish removal raises Italian and Portuguese perplexity (Zhang et al. 2024, Table 11), and deactivating French neurons also affects Spanish (Tang et al. 2024, sec. 3.3.1).

Parameter decompositions may allow narrower edits. adVersarial Parameter Decomposition (VPD)[1] approximates each weight matrix of a model by a sum of rank-one *subcomponents*, trained so that few are needed on any input (Bushnaq et al. 2026, sec. 2). It follows earlier attribution-based and stochastic decompositions (Braun et al. 2025; Bushnaq et al. 2025). If a subcomponent is specific to one language, an edit along it, $W \leftarrow W + \alpha A$, might change that language and spare its relatives. This study uses Goodfire's four-layer, 67M-parameter decoder-only model decomposed in the VPD report.

This study is a small follow-up inspired by Lucius Bushnaq's LessWrong post, "Exploration: fine-tuning with parameter decomposition". Bushnaq trains the scalar prefactors of selected subcomponents of the same model to suppress German. One subcomponent, chosen by its interpretation label, reaches chance-level German with little English damage and spares French and Spanish, while also damaging Italian (Bushnaq 2026). His comparison uses matched training budgets and LoRA baselines, with varying achieved effects.

I extend Bushnaq's scalar-edit comparison to three language pairs with a fixed rule for selecting one subcomponent per target from target and English text. I match target effects on validation text and add full-matrix fine-tuning and neuron baselines.

[1] A separate experiment in the VPD report tests manual changes to a subcomponent's write vector using an unembedding direction. A LoRA update of the same matrix trained on 947 examples does better on both of its KL measures (Bushnaq et al. 2026, sec. 6).

The pairs are German (de) and Dutch (nl), Spanish (es) and Portuguese (pt), and French (fr) and Italian (it). In each pair the related language is *protected*: it is evaluated after selection and training. I calibrate each edit to raise target cross-entropy (CE) by $\tau = 0.5$ or 1 nat/token on validation text.

The outcome is the CE rise of the protected language on European Parliament session text (Koehn 2005). After fixing the six edits and comparators, I evaluate them once on 58 new session dates to test whether the difference in related-language damage carries over to new text.

2 Setup

The method details give window and set sizes, grids, tolerances and training recipes.

Model and subcomponents. The base model W_0 is the four-layer, 67M-parameter decoder-only model decomposed in the VPD report. It was trained on a subset of the Pile (Bushnaq et al. 2026, sec. 3.1; Gao et al. 2020). Edits act on its 24 weight matrices: in each layer, the query, key, value and output projections of attention and the two MLP matrices. The decomposition supplies 38,912 rank-one subcomponents in all, each of the form $A_c = u_c v_c^\top$. Subcomponents are named by layer, matrix and index: `0.attn.o:268` is subcomponent 268 of the attention output projection of layer 0.

The edit. An edit along subcomponent c of matrix m is $W_m \leftarrow W_m + \alpha A_c$, with the subcomponent not normalized. In VPD every subcomponent carries a mask coefficient that equals 1 in the unedited model. The edit moves it to $1 + \alpha$. We write $\alpha = s \cdot a$, with a sign $s = \pm 1$ and a dose $a > 0$. With $s = -1$, as for every subcomponent selected here, the coefficient is $1 - a$: a dose below 1 shrinks the subcomponent's contribution, a dose of 1 removes it, and a dose above 1 reverses its sign. The edit norm is $|\alpha| \, \|A_c\|_F$.

Text and effects. Every Europarl set has one window per session date, parallel across languages. The *CE rise* of an edit is measured in nat/token. For each window, I subtract the base model W_0's mean next-token CE from the edited model's, then average these differences over windows. Each target has three overlapping *draws* of its Europarl training windows. The English *retain pool* mixes Europarl English training text, WikiText-2 validation text (Merity et al. 2017) and English-filtered Pile validation text.

2.1 Selecting the subcomponent

The rule scores each subcomponent by its local effect on target CE relative to its local effect on English output. Let g_c be the derivative of the mean target CE on the draw with respect to the subcomponent's mask coefficient. Let q_c be the curvature of the English output along the same coordinate, estimated on the retain pool with labels sampled from W_0. This estimates the directional Fisher information of the model's predictive distribution. The rule ranks subcomponents by the *gradient-to-retain-curvature score*

$$\frac{g_c^2}{q_c + \varepsilon \|A_c\|^2},$$

where a small floor ε keeps subcomponents with almost no measured curvature from dominating.

The rule then tests a shortlist of the best-ranked subcomponents on the target training draw. For each subcomponent, with sign $s = \operatorname{sign}(g_c)$, it raises the dose step by step up to 2. It accepts the first subcomponent whose target CE rise reaches 1 nat/token. If none does, it accepts the first that reached 0.5, and otherwise it finds no feasible subcomponent.

The rule chose the same subcomponent in all three draws of each target: `0.attn.o:268` for German, `0.attn.o:676` for Spanish and `0.attn.o:640` for French. They all lie in the attention output projection of the first layer and have sign -1.

2.2 Comparators

Like the selection rule, the comparators use only target and English text for selection and training.

- **Rank-one LoRA** (Hu et al. 2022) learns both vectors of an update to the matrix that holds the selected subcomponent, the attention output projection of layer 0. It uses the same matrix and rank as the subcomponent edit.
- **Full-matrix fine-tuning** updates all 24 weight matrices.
- **Neuron edits** use the top k MLP neurons ranked by how much more often they are active on the target draw than on the English retain pool. The edit scales their output weights down, with k and the scale calibrated to τ, in the spirit of language-specific neurons (Tang et al. 2024; Kojima et al. 2024).
- **The randomized-dictionary control** applies the same selection rule to a randomized dictionary. In each matrix, one random signed permutation transforms the output coordinates of every subcomponent, and another transforms the input coordinates. This preserves their norms and pairwise angles but removes their alignment with W_0.

The two learned updates are trained, one per draw, on a target CE objective plus the KL divergence from W_0 on the English retain pool, for a fixed 200 steps. The final update is scaled down to reach τ.

A separate comparison family uses single subcomponents chosen by alternative selection rules:

- the top subcomponent of an orthogonal matching pursuit (Pati et al. 1993; Mallat and Zhang 1993) of the full-matrix fine-tuning update or of a LoRA update trained across all matrices
- a causal-importance heuristic that takes the top-1 subcomponent, adapted from Bushnaq's top-16 procedure (Bushnaq 2026)
- for German only, the previously chosen subcomponent `0.attn.o:268`.

These comparisons have their own multiplicity correction and receive no confirmatory verdicts (appendix).

2.3 Calibration and the six edits

Every method is calibrated on target Europarl validation text. A short search over its dose sets the target CE rise to τ. An edit is *matched* if its validation effect lies within 0.02 nat/token of τ. Only matched edits enter a contrast. The effects achieved on the readout dates can differ and are reported with the results.

Table 1: The six VPD subcomponent interventions, each shared by the three draws: dose a, the subcomponent's mask coefficient $1 - a$ after the edit, edit norm $|\alpha| \cdot \|A\|$ and the achieved CE rises in nat/token. C: confirmation. D: development.

Pair	Subcomponent	τ	Dose a	Coefficient	Edit norm	Target rise, D	Target rise, C	Protected rise, C
de → nl	0.attn.o:268	0.5	0.681	0.319	0.29	0.49	0.48	0.0030
		1	0.945	0.055	0.40	1.00	0.98	0.0053
es → pt	0.attn.o:676	0.5	1.026	−0.026	0.34	0.53	0.52	0.0003
		1	1.388	−0.388	0.47	1.05	1.02	0.0030
fr → it	0.attn.o:640	0.5	0.762	0.238	0.31	0.53	0.50	0.0008
		1	1.039	−0.039	0.42	1.06	1.01	0.0017

The three draws yield the same six fixed scaled edits, one per target and τ. Three reduce the selected contribution without reversing it, and three reverse its sign. All coefficients remain nonzero (Table 1).

2.4 What was fixed, and what was known

Selection and training used only target and English text. Earlier research had used protected-language development text and identified the German subcomponent, so the selection history was not blind (Section A.3).

I fixed the edits, comparators and analysis before evaluating them once on 58 further session dates. Those dates were not used for selection, calibration or re-matching. This tests the six fixed edits on new text in the same languages. Evaluating the selection procedure itself remains an open question.

2.5 Readouts, eligibility and intervals

Guards. English test sets determine which matched edits are usable: Europarl, WikiText-2 and Pile test text on the development readout, and English text from the confirmatory dates on the confirmatory readout. An edit is unusable if any guard's CE rises by more than 0.1 nat/token. On the confirmatory dates it must also have been usable on development. This is an eligibility rule fixed in advance. Two usable edits can still differ greatly in how much they change English (Section 3.4).

Contrasts. For each protected-language session date, a contrast takes the CE rise of the subcomponent edit minus that of the comparator, averaged over the draws in which both are matched and usable. Its estimate is the mean over dates. Negative values mean less protected-language damage for the subcomponent edit. The *claim family* comprises four comparators $\times$ three pairs $\times$ two values of τ, or 24 planned contrasts.

Intervals. Every confidence interval in this paper is a simultaneous 95% interval from a bootstrap over session dates (Efron 1979), with one resample shared by every contrast of a family and the max-|t| critical value of a simultaneous band (Montiel Olea and Plagborg-Møller 2019, supplement, Algorithm 3). The intervals describe uncertainty across session dates for these fixed edits. The bootstrap varies dates while holding the draws, training seeds, subcomponents, language pairs and model fixed.

Confirmatory verdicts. Each claim contrast with a development interval receives one verdict on the confirmatory readout, according to a rule table fixed before the set was opened. It is "reproduced on confirm" if its interval excludes zero with the development sign and no used draw has a target CE rise outside $\tau \pm 0.1$. A rise outside this band is the *target flag*.

Post hoc analyses. Two analyses were chosen after the results were known, outside the protocol. The frozen verdicts remain those of the original analysis. The first asks whether the target effects achieved on the readout dates were comparable (Section 3.3). The second counts how often the learned updates move English less (Section 3.4).

3 Results

3.1 Target change against protected damage

All matched edits passed the English guards, and on the confirmatory dates no edit raised the target flag. Figure 1 shows each edit on the confirmatory dates as target CE rise against protected-language CE rise. At both values of τ and for all three pairs, the subcomponent edits have the lowest protected-language CE rise. Across the six subcomponent edits, the protected-language rise is 0.0003 to 0.0053 nat/token. The other methods raise it by about 0.03 to 0.88 (Table 2).

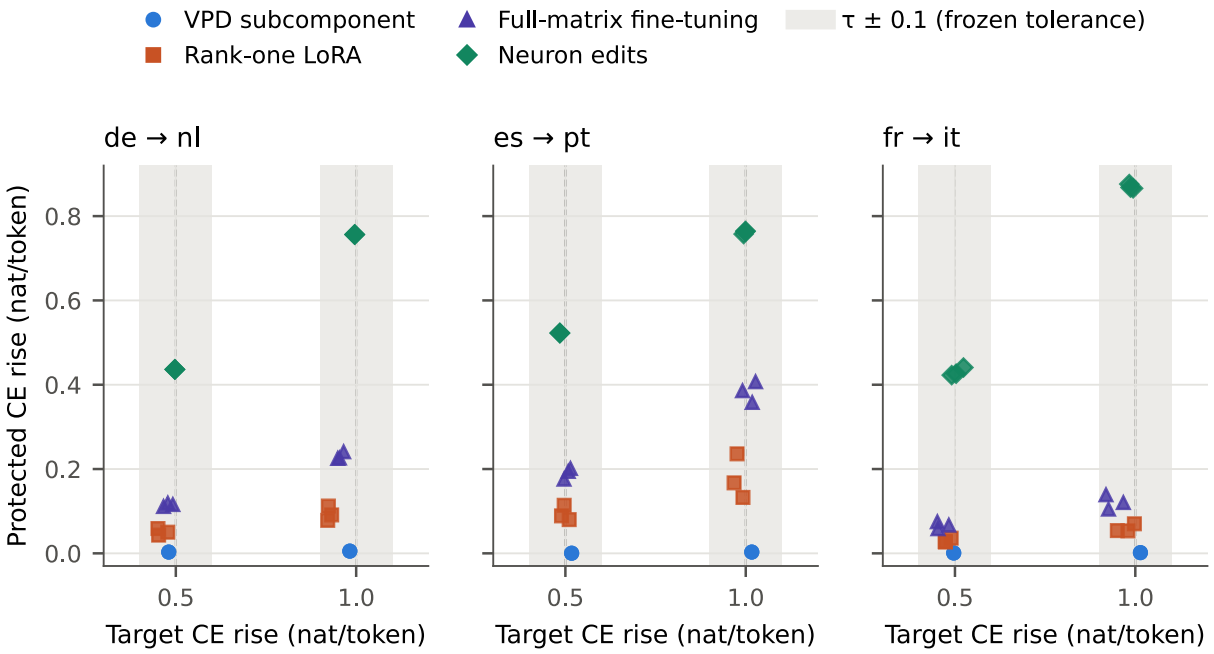


Figure 1: Target CE rise against protected CE rise for every edit on the 58 confirmatory dates (subcomponent edits: one per pair and τ, other methods: one per draw). The bands mark τ ± 0.1, the frozen tolerance of the target flag. At both values of τ the subcomponent edits raise the protected language's CE far less than the edits calibrated to the same τ.

3.2 The protected contrasts

On the confirmatory dates, all 18 claim contrasts that have an estimate lie below zero, with upper interval ends at −0.023 or below, and every verdict is "reproduced on confirm" (Figure 2, top row, Table 4). The largest difference between a confirmatory estimate and its development estimate is 0.033.

The 18 contrasts share six subcomponent edits: each edit is compared with three comparators. The three draws vary only the comparators' training data, so the contrasts are dependent.

3.3 Achieved target effects

The edits were calibrated to the same effect on validation text. On the readout dates the achieved effects differ somewhat (Figure 1). To assess whether less protected damage came with a smaller target effect, I estimated, post hoc, the difference in target CE rise between the subcomponent edit and each comparator, its *target gap*, in one simultaneous family with the protected contrasts (method details). A target gap counts as agreement if its whole interval lies within ±0.1 nat/token. This band was chosen after the results. It compares two methods and is separate from the frozen target flag, which compares one edit with τ.

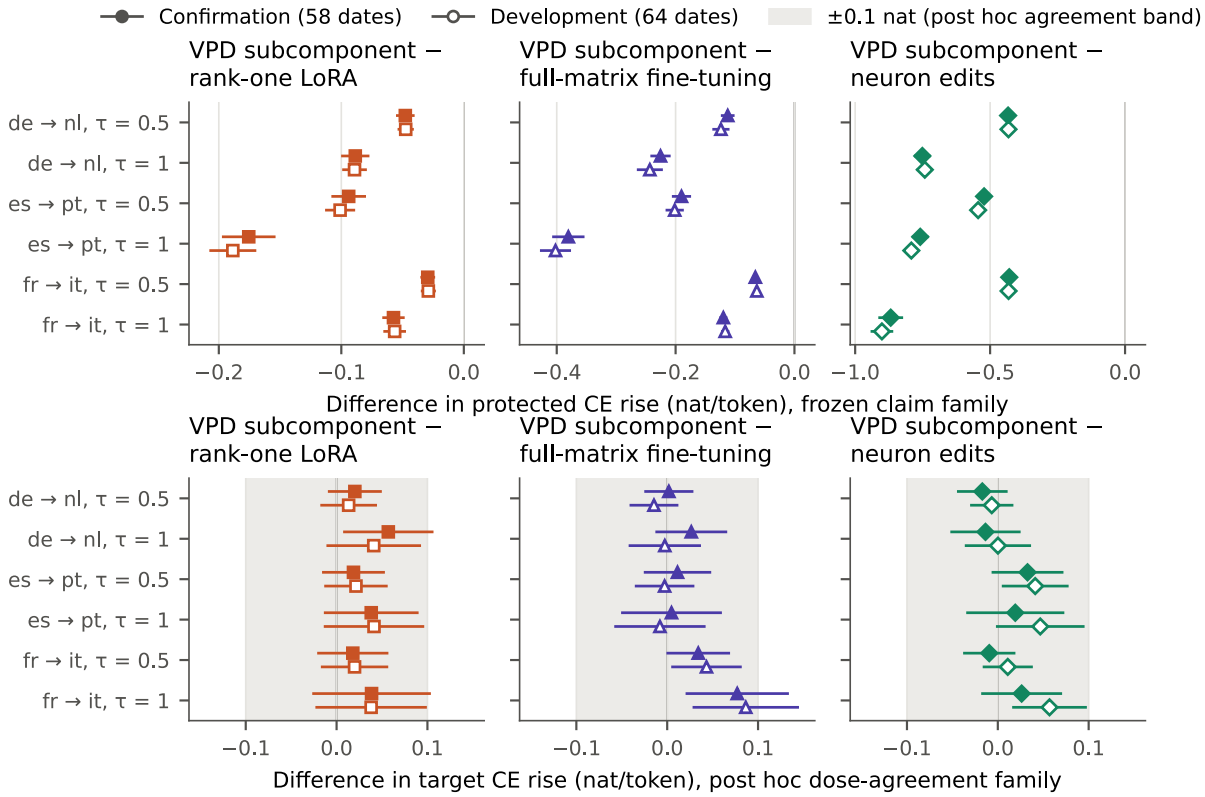


Figure 2: Contrasts of the subcomponent edit with each claim comparator, on the confirmatory (filled, upper marker) and development (open, lower marker) dates, with simultaneous 95% intervals. Top: difference in protected CE rise, the frozen claim family. The axes differ between columns. Bottom: difference in achieved target CE rise, from the post hoc family of Section 3.3, with the ±0.1 nat/token agreement band. Negative values in the top row mean less protected-language damage for the subcomponent edit. Positive values in the bottom row mean a larger target effect.

On the confirmatory dates, 15 of the 18 target gaps agree by this criterion (Figure 2, bottom row). The three exceptions are all at $\tau = 1$, and in each the point estimate lies inside the band:

- German and Dutch against rank-one LoRA: +0.057 [+0.007, +0.107]
- French and Italian against full-matrix fine-tuning: +0.077 [+0.020, +0.134]
- French and Italian against rank-one LoRA: +0.038 [−0.027, +0.104].

On the development dates, 17 of 18 target gaps agree. The exception is French and Italian against full-matrix fine-tuning. All four exceptions have positive point estimates, though the interval for French and Italian against rank-one LoRA permits a smaller target effect for the subcomponent edit. In this larger post hoc family, all 18 protected-CE intervals and all 18 protected-KL intervals lie below zero on both readouts. Protected KL is $D_{\mathrm{KL}}(p_{W_0} \,\|\, p_{\mathrm{edit}})$, averaged over next-token positions on protected-language text.

3.4 English: where the learned updates are better

The guard rule only asks whether English CE rose past its limit. A finer measure is the KL divergence $D_{\mathrm{KL}}(p_{W_0} \,\|\, p_{\mathrm{edit}})$ of the edited model's English next-token distribution from that of W_0, averaged over positions. By this measure, both learned updates preserve English better than the subcomponent edits (Table 2). On the development readout, over three pairs × three draws × two values of τ × four English sets (the validation set and the three guards), the LoRA update has the lower KL in 72 of 72 comparisons and full-matrix fine-tuning in 65 of 72. On the English confirmatory text both have the lower KL in 18 of 18.

The subcomponent edits also change English little in absolute terms: their English KL on the confirmatory dates is below 10^{-3} nat/token. Their weight changes have larger norms: 0.29 to 0.47, against 0.12 to 0.24 for the LoRA updates and 0.07 to 0.11 for full-matrix fine-tuning. The learned updates were trained with an explicit

English KL term. None of the methods had a term for the protected language. The comparison therefore measures how well the methods spare a related language absent from their training objective.

Table 2: What the edits did on the 58 confirmation dates, as the range over 9 edits per row (3 targets × 3 draws, with the VPD subcomponent draws sharing one edit per target). Rises are CE rises in nat/token on the target, the protected language and English. C: confirmation. D: development. English KL is from the base model. On development data it covers the English validation set and the three guards. Edit norm: Frobenius norm of the weight change.

Method	τ	Target rise	Protected rise	English rise	English KL, C (10^{-3})	English KL, D (10^{-3})	Edit norm
VPD subcomponent	0.5	0.48–0.52	0.0003–0.0030	0.0002–0.0012	0.28–0.41	0.17–0.56	0.29–0.34
	1	0.98–1.02	0.0017–0.0053	0.0003–0.0019	0.49–0.77	0.31–1.01	0.40–0.47
Rank-one LoRA	0.5	0.45–0.51	0.027–0.114	−0.0002 to 0.0001	0.05–0.08	0.04–0.16	0.12–0.17
	1	0.92–1.00	0.053–0.236	−0.0004 to 0.0001	0.08–0.15	0.08–0.25	0.15–0.24
Full-matrix fine-tuning	0.5	0.45–0.51	0.058–0.201	0.0007–0.0014	0.08–0.21	0.08–0.24	0.07–0.08
	1	0.92–1.03	0.105–0.407	0.0010–0.0020	0.16–0.42	0.15–0.49	0.10–0.11
Neuron edits	0.5	0.48–0.52	0.423–0.523	−0.0008 to 0.0010	3.29–5.63	0.61–5.45	1.12–3.16
	1	0.98–1.00	0.756–0.876	0.013–0.038	13.38–38.67	11.85–39.86	5.85–9.80

3.5 The randomized-dictionary control gave no estimate

The control applies the selection rule to the randomized dictionary. It found no feasible subcomponent in any of the 9 target–draw combinations: no shortlisted subcomponent reached even 0.5 nat/token on the target training draw within the dose limit of 2. The randomized-dictionary control therefore has no edit, and its six claim contrasts have no estimate. Because some planned contrasts are empty, the frozen analysis has the overall status “conditional” rather than “complete”.

The control was meant to measure how selective an edit of the same form is when the subcomponents are unrelated to the model’s weights. Its failure to produce an edit leaves the dictionary’s contribution unresolved. Whether a randomized subcomponent at a larger dose could spare the protected language remains unknown. The alternative subcomponent comparisons and whitened rank-one selector leave the contributions of the dictionary and selection rule unresolved as well (Section 3.6; Section 4).

3.6 Subcomponents chosen by other rules

The alternative selection comparisons use single subcomponents of the same dictionary. Every estimable contrast is at or below zero. The causal-importance heuristic picked `0.attn.o:41` in 8 of 9 target–draw combinations and, on the confirmatory dates, raised the protected CE by 0.48 to 1.37 nat/token more than the selected subcomponent, at achieved target rises close to the selected subcomponent’s (Table 5). The previously chosen German subcomponent is the selected one, so it gives an identical edit. Selectivity therefore varies among subcomponents of the same dictionary. By contrast, the randomized-dictionary control held the selection rule fixed and changed the dictionary (Section 3.5).

4 Longer training and alternative rank-one directions

After the confirmatory readout, I ran exploratory comparisons on the previously evaluated development dates. The six subcomponent edits were compared with a whitened rank-one selector outside the VPD dictionary and learned updates trained for 2,000 steps, all calibrated to the same τ. All 18 protected-CE intervals of the all-run family and all 12 of the family excluding unstable learned runs lie below zero. The achieved target rises differ by at most 0.07 nat/token. The appendix gives the methods, full results and study record.

The expanded training recipe lowered the protected-language rise of full-matrix fine-tuning in all six target and τ combinations and that of LoRA in five of six. Steps, margin, grid and seeds changed together, with retention still limited to English. Convergence remains unestablished at either budget.

The whitened rank-one selector uses the selected VPD subcomponent's norm and the main selector's ε, leaving the dictionary's contribution entangled with the selector's. The SVD comparator gave no estimate.

5 Limits of these conclusions

The result covers six edits in one four-layer model, three language pairs and two values of τ. The measured outcomes are protected-language CE on parliamentary text and English CE guards and KL. The confirmatory readout tests fixed edits on new text in the same languages; uncertainty varies session dates only (Section 2.5). Testing the selection rule in other models, explaining the edits' selectivity and evaluating tasks beyond next-token prediction remain open.

6 Conclusion

In this model, one subcomponent per target language, chosen from target and English text and calibrated to raise target CE by 0.5 or 1 nat/token on validation text, raised the CE of the related language by at most 0.0053 nat/token on 58 new session dates. Rank-one LoRA and full-matrix fine-tuning updates calibrated to the same validation rise raised it by about 0.03 to 0.41, and neuron edits by 0.42 to 0.88. All 18 simultaneous intervals of the differences lie below zero. A post hoc check found agreement in achieved target effects within ± 0.1 nat/token in 15 of the 18 comparisons. In the other three the subcomponent edit's target effect was larger on average, though one interval also allows a smaller effect. The advantage does not extend to English, where the learned updates move the output distribution less.

Exploratory comparisons on the development dates found the same direction against a whitened rank-one selector and against learned updates trained for 2,000 steps. The randomized-dictionary control found no edit to compare, and the selection history was not blind (Section 2.4). Together, the comparisons show that selective single-subcomponent edits exist for these three pairs in this model, with effects reproduced on a fixed confirmatory readout.

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A Method details

Subcomponents and text. The decomposition has 512 subcomponents each for query and key, 1,024 each for value and output, 3,072 for the MLP input and 3,584 for the MLP output matrix of each layer. The Europarl text comes from release v7 of the corpus (Koehn 2005), downloaded from https://www.statmt.org/europarl/v7/europarl.tgz. Each Europarl split has one 257-token window per session date (256 predictions). Each target has three draws of 64 of its 128 Europarl training windows. Calibration uses 32 target-language Europarl validation windows per target. The English validation set has 64 windows: 32 Europarl and 32 English-filtered Pile windows. The English retain pool has 256 windows: 128 of Europarl English training text, 64 of WikiText-2 validation text and 64 of English-filtered Pile validation text.

Selection. For softmax cross-entropy, the Fisher information matrix equals the generalized Gauss–Newton matrix (Martens 2020, sec. 9.2). It differs from the empirical Fisher, which uses the data's labels (Kunstner et al. 2019, sec. 1). The curvature q_c is the mean squared directional score over the retain pool, with eight sets of labels sampled from W_0 per window. The floor ε is the 1st percentile of $q_c/\|A_c\|^2$ over subcomponents. The top 64 subcomponents are re-ranked with the larger of q_c and a second estimate on 256 other English windows. The search raises the dose on the grid $0.0625 \cdot \sqrt{2}^{\,k}$ for $k = 0, \ldots, 10$, that is up to 2. Before any selection, q was checked against finite-difference curvature on English windows. The check, including contraction, batch splitting and scaling, passed on the NVIDIA L40S used for the original runs. On the tested A10 and L4, the batch-splitting test failed identically in the original and released implementations. The checked selection reruns are limited to the L40S, for the German target's first draw with the VPD dictionary and one randomized dictionary. Fisher-label sampling depends on the GPU model, so selections can differ on other GPUs.

Calibration. The dose grid is scanned and the first crossing of τ is bracketed. Linear bisection, at most five steps, stops at the first point within the matching tolerance. The neuron edit scales the output weights of the top k neurons by $1 - f$. Its calibration takes the smallest k among powers of two from 1 to 256 whose full removal reaches τ, then calibrates f. The chosen k ranges from 2 to 256. The randomized-dictionary permutations use a separate seed per draw.

Learned updates. The objective is a hinge on target CE, $\max(0, 1 - \Delta\text{CE})$ per window, plus the KL divergence from W_0 on the English retain pool with weight 1. Training uses Adam (Kingma and Ba 2015) with 10 warm-up steps and gradient clipping at 1.0, 8 target and 8 English windows per step, for 200 steps. The edit is the final update D scaled down, βD with $\beta \leq 1$, and β is calibrated like a dose. The learning rate was chosen once per target, on draw 0, by the lowest English validation KL at $\tau = 1$. The LoRA grid was {1e-4, 3e-4, 1e-3, 3e-3} and the full-matrix fine-tuning grid {1e-6, 3e-6, 1e-5}. The choice was 3e-3 for LoRA and 1e-6 for full-matrix fine-tuning on all three targets, the top and the bottom of their grids.

A.1 Extended comparisons

The comparisons were specified before these runs. I measured the six subcomponent edits again, next to four comparators calibrated to the same τ:

- **SVD components.** The selection rule of Section 2.1 is applied unchanged to the rank-one terms of a singular value decomposition of the 24 weight matrices. These terms reconstruct the weights, whereas VPD leaves a residual that is unchanged by its subcomponent edits.
- **A whitened rank-one selector.** In each matrix, this procedure finds a direction outside the VPD dictionary with the largest squared target gradient relative to the damped K-FAC curvature of the English retain text (Martens and Grosse 2015), using the norm of that target–draw combination's VPD subcomponent and the selector's ε. It then scores the direction with the same rule and tests doses from 1/64 to 8.
- **Longer-trained rank-one LoRA and full-matrix fine-tuning.** The learned updates are trained for 2,000 steps instead of 200, with a hinge margin of 1.5 instead of 1, wider learning-rate grids and three seeds per draw.

Each side of a contrast is the mean over its matched and usable runs. The analysis compares these means without pairing runs by draw. The contrasts form one simultaneous family. The learned-update contrasts are repeated as a second family without the runs flagged as unstable, 7 of the 27 LoRA runs. For each interval, the analysis reports its direction and whether it lies wholly within ± 0.02 nat/token. These runs used a different GPU model, which changes the sampled Fisher labels. Selection found the same subcomponents in all nine target–draw combinations, and the subcomponent-edit readout matched the main readout within $1.5 \cdot 10^{-6}$ nat/token.

Table 3: Exploratory comparisons on the 64 development dates: VPD subcomponent minus comparator, difference in protected CE rise in nat/token, with simultaneous 95% intervals. All runs and stable runs (learned updates without unstable runs) are separate families. Every interval lies below zero and none lies within ±0.02, in both families. Runs: the comparator's matched and usable runs (draws, times three seeds for the learned updates), and in parentheses those that are stable. Each contrast includes all three VPD subcomponent draws. Target rise: the achieved target-test CE rise, mean over all runs, and in parentheses over stable runs where it differs.

Pair	τ	Comparator	Runs (stable)	Protected, all runs	Protected, stable runs	Target rise, subcomponent / comparator
de → nl	0.5	Whitened rank-one selection	3	−0.076 [−0.084, −0.068]		0.49 / 0.50
de → nl	0.5	Rank-one LoRA, longer training	9 (8)	−0.044 [−0.050, −0.038]	−0.044 [−0.050, −0.038]	0.49 / 0.49
de → nl	0.5	Full-matrix fine-tuning, longer training	9 (9)	−0.098 [−0.111, −0.086]	−0.098 [−0.111, −0.086]	0.49 / 0.51
de → nl	0.5	SVD component	0 of 3	no estimate: 3 failed the guard		
de → nl	1	Whitened rank-one selection	3	−0.170 [−0.183, −0.157]		1.00 / 0.98
de → nl	1	Rank-one LoRA, longer training	9 (8)	−0.073 [−0.081, −0.064]	−0.072 [−0.080, −0.064]	1.00 / 0.96
de → nl	1	Full-matrix fine-tuning, longer training	9 (9)	−0.181 [−0.200, −0.163]	−0.181 [−0.199, −0.164]	1.00 / 0.99
de → nl	1	SVD component	0 of 3	no estimate: 3 failed the guard		
es → pt	0.5	Whitened rank-one selection	3	−0.201 [−0.216, −0.186]		0.53 / 0.49
es → pt	0.5	Rank-one LoRA, longer training	9 (6)	−0.105 [−0.117, −0.092]	−0.110 [−0.122, −0.097]	0.53 / 0.50
es → pt	0.5	Full-matrix fine-tuning, longer training	9 (9)	−0.170 [−0.186, −0.155]	−0.170 [−0.185, −0.156]	0.53 / 0.52
es → pt	0.5	SVD component	0 of 3	no estimate: 3 failed the guard		
es → pt	1	Whitened rank-one selection	3	−0.442 [−0.471, −0.414]		1.05 / 0.99
es → pt	1	Rank-one LoRA, longer training	9 (6)	−0.183 [−0.202, −0.164]	−0.195 [−0.214, −0.176]	1.05 / 1.02
es → pt	1	Full-matrix fine-tuning, longer training	9 (9)	−0.337 [−0.363, −0.310]	−0.337 [−0.363, −0.311]	1.05 / 1.04
es → pt	1	SVD component	0 of 3	no estimate: 2 not reached, 1 failed the guard		
fr → it	0.5	Whitened rank-one selection	3	−0.027 [−0.031, −0.023]		0.53 / 0.49
fr → it	0.5	Rank-one LoRA, longer training	9 (6)	−0.017 [−0.021, −0.013]	−0.017 [−0.021, −0.014]	0.53 / 0.54
fr → it	0.5	Full-matrix fine-tuning, longer training	9 (9)	−0.039 [−0.043, −0.035]	−0.039 [−0.043, −0.035]	0.53 / 0.49
fr → it	0.5	SVD component	0 of 3	no estimate: 3 failed the guard		
fr → it	1	Whitened rank-one selection	3	−0.050 [−0.056, −0.045]		1.06 / 0.99
fr → it	1	Rank-one LoRA, longer training	9 (6)	−0.027 [−0.032, −0.021]	−0.027 [−0.032, −0.022]	1.06 / 1.08 (1.07)

Pair	τ	Comparator	Runs (stable)	Protected, all runs	Protected, stable runs	Target rise, subcomponent / comparator
fr → it	1	Full-matrix fine-tuning, longer training	9 (9)	−0.070 [−0.076, −0.065]	−0.070 [−0.075, −0.065]	1.06 / 1.00
fr → it	1	SVD component	0 of 3	no estimate: 2 failed the guard, 1 not reached		

All 18 estimable intervals of the all-run family and all 12 of the stable-run family lie below zero, and none lies within ±0.02 (Table 3). The closest to the band is French and Italian at $\tau = 0.5$ against rank-one LoRA, −0.017 [−0.021, −0.013]. The achieved target rises of the two sides differ by at most 0.07 nat/token. The subcomponent edit's target rise is larger in 14 of 18 comparisons.

The longer budget lowered the protected-language rise of full-matrix fine-tuning in all six target and τ combinations, to 0.04–0.34 nat/token from 0.06–0.40 at 200 steps, and that of LoRA in five of six, to 0.02–0.18 from 0.03–0.19. The SVD comparator gave no estimate: every SVD edit that reached τ raised an English guard by more than 0.1 nat/token, and three of nine did not reach $\tau = 1$.

Chronology of the exploratory comparisons. The comparisons, labels and limits were declared on 12 October 2026, before the runs. The 5% tolerance for reproducing the main selection's curvature floor across GPU models was added after one selection had finished but before its result was read. The analysis rules were fixed before any readout, after the selection outcomes, learning-rate choices and target and English validation effects of the rate grid had been seen. The records were then frozen, and the readout was run. The analyzer's two properties were implemented after readout but before any readout measurements were opened or analysed. This sequence comes from the author's records and is not independently timestamped.

Diagnostics for the exploratory comparisons. A learned run is flagged as unstable if its English validation KL at step 2,000 is more than twice its minimum from step 100 onward. The second comparison family excludes these runs. A change of less than 10% in English validation KL between steps 1,500 and 2,000 is called a plateau. Only 6 of the 54 training runs used in the comparisons met it. Including the 18 additional learning-rate candidates gives 7 of 72, all full-matrix fine-tuning runs. This diagnostic does not establish convergence of the training objective and does not determine which edit is selected. At 200 steps the learned comparators' learning rates lay at the edges of the tested grids. At 2,000 steps they still caused more related-language damage.

A.2 Readouts and intervals

Readouts. The development guards are Europarl English test text (64 windows), WikiText-2 test text (32 windows) and Pile test documents (512 windows). The confirmatory guard is English text from the 58 confirmatory dates. The confirmatory set has 48 main and 10 reserve dates, analysed as one set.

Intervals. Dates are resampled with replacement, one resample shared by every contrast of a family. Each contrast's standard error is the standard deviation of its resampled means, and the interval is the estimate $\pm\, c$ standard errors. The critical value c is the 95% quantile, over resamples, of the largest absolute standardized deviation in the family. With 2,000 resamples (seed 0), $c = 2.780$ for the claim family on the 58 confirmatory dates and 2.814 on the 64 development dates. For the alternative subcomponent-selection family, c is 2.817 on the confirmatory dates and 2.772 on development.

Verdicts. The first applicable rule in the verdict table determines the result. No interval or a technical failure gives "not established". A target flag in any used draw gives "effect not maintained". An interval that excludes zero with the development sign gives "reproduced on confirm", and with the opposite sign "not reproduced". Anything else is "not established".

Target gaps. The post hoc target gaps, the protected-CE contrasts and the corresponding protected-KL contrasts form one family of 54 coordinates per readout, with the per-date values averaged over draws. The family has a joint bootstrap over dates (20,000 resamples, seed 20261011) and a max-|t| critical value of 3.149 on the confirmatory readout and 3.158 on the development dates.

A.3 Protocol records and prior access

- **Freeze and readout.** The selection rule, comparators, calibration, guard rule and analysis were fixed in code and a protocol, with a freeze record dated 11 October 2026. The development readout and its

analysis ran right after the freeze. A second set of 58 session dates was read once, later the same day, with the frozen edits unchanged. An amendment written before any window was parsed fixed the readout procedure.

- **Prior knowledge.** The research before the freeze was not blind. The German subcomponent `0.attn.o:268` had been chosen in an earlier experiment with Dutch in view, and it was the top German candidate in an earlier pilot run of this selector. Dutch, Portuguese and Italian text from the development splits, and the NTREX Dutch and Italian sets (Federmann et al. 2022), had been read or used in earlier development. The protected-language damage on the 64 development dates is therefore part of the development readout.
- **Pretraining.** The Pile contains a European Parliament component (Gao et al. 2020, sec. 2.17). I did not establish whether the subset on which the model was trained includes it, so the dates are new to this workflow but may have appeared in pretraining.
- **Dates and hashes.** The dates come from the documents' own records and the project's run log, and neither is independently timestamped. The original protocol, freeze record and frozen code list are not published. Their recorded hashes identify the files' contents; the date of those contents remains unverified, so the records lack a timestamped preregistration.

B The claim contrasts

Table 4 lists the 18 claim contrasts of Figure 2 with their intervals. The six randomized-dictionary contrasts have no estimate (Section 3.5).

Table 4: The 18 claim contrasts, VPD subcomponent minus comparator, in nat/token, with simultaneous 95% intervals. Protected: difference in protected CE rise, the frozen claim family. Target gap: difference in achieved target CE rise, from the post hoc 54-coordinate family.

Pair	τ	Comparator	Protected, confirmation	Protected, development	Target gap, confirmation
de → nl	0.5	Rank-one LoRA	−0.048 [−0.055, −0.040]	−0.047 [−0.054, −0.041]	+0.020 [−0.010, +0.050]
		Full-matrix fine-tuning	−0.112 [−0.124, −0.101]	−0.124 [−0.138, −0.109]	+0.002 [−0.025, +0.029]
		Neuron edits	−0.433 [−0.457, −0.410]	−0.432 [−0.456, −0.408]	−0.017 [−0.045, +0.011]
	1	Rank-one LoRA	−0.089 [−0.100, −0.077]	−0.089 [−0.099, −0.079]	+0.057 [+0.007, +0.107]
		Full-matrix fine-tuning	−0.225 [−0.242, −0.208]	−0.243 [−0.265, −0.221]	+0.027 [−0.013, +0.066]
		Neuron edits	−0.751 [−0.782, −0.720]	−0.742 [−0.772, −0.712]	−0.014 [−0.052, +0.025]
es → pt	0.5	Rank-one LoRA	−0.094 [−0.108, −0.080]	−0.101 [−0.113, −0.089]	+0.019 [−0.016, +0.053]
		Full-matrix fine-tuning	−0.190 [−0.206, −0.174]	−0.202 [−0.217, −0.186]	+0.011 [−0.026, +0.048]
		Neuron edits	−0.522 [−0.549, −0.496]	−0.544 [−0.570, −0.519]	+0.033 [−0.007, +0.072]
	1	Rank-one LoRA	−0.176 [−0.197, −0.154]	−0.189 [−0.208, −0.169]	+0.038 [−0.014, +0.090]
		Full-matrix fine-tuning	−0.380 [−0.408, −0.353]	−0.402 [−0.428, −0.376]	+0.005 [−0.051, +0.060]
		Neuron edits	−0.759 [−0.790, −0.728]	−0.792 [−0.821, −0.763]	+0.019 [−0.035, +0.073]
fr → it	0.5	Rank-one LoRA	−0.029 [−0.035, −0.023]	−0.029 [−0.035, −0.023]	+0.018 [−0.021, +0.057]
		Full-matrix fine-tuning	−0.066 [−0.071, −0.061]	−0.063 [−0.068, −0.058]	+0.034 [−0.001, +0.069]
		Neuron edits	−0.429 [−0.456, −0.402]	−0.432 [−0.456, −0.407]	−0.009 [−0.038, +0.019]
	1	Rank-one LoRA	−0.057 [−0.067, −0.048]	−0.056 [−0.066, −0.047]	+0.038 [−0.027, +0.104]
		Full-matrix fine-tuning	−0.120 [−0.126, −0.113]	−0.117 [−0.123, −0.110]	+0.077 [+0.020, +0.134]
		Neuron edits	−0.868 [−0.914, −0.822]	−0.901 [−0.943, −0.859]	+0.026 [−0.018, +0.071]

C Alternative subcomponent-selection comparisons

Each alternative selection rule chooses one subcomponent, which is calibrated in the same way as the main edit:

- `omp_finetune` takes the first subcomponent of an orthogonal matching pursuit of the chosen full-matrix fine-tuning update, the subcomponent with the largest absolute correlation with it, with the sign of that correlation.
- `omp_lora` does the same for a rank-one LoRA on all 24 matrices, learning rate 1e-4, with a checkpoint rule that uses English text only for retention. This differs from the main rank-one LoRA comparator, which updates one matrix.
- `ci_difference` is a causal-importance heuristic that takes the top-1 subcomponent: the subcomponent with the largest mean causal importance on the target draw minus that on the English retain pool, on the removal side. Causal importance is the per-position importance learned for each subcomponent in VPD (Bushnaq et al. 2026, sec. 2.3). Bushnaq ranks subcomponents the same way but trains the top 16, and his single-subcomponent edit used a subcomponent chosen by its label instead of the ranking (Bushnaq 2026). The selection heuristic adapts that top-16 procedure to selecting one subcomponent.
- `historical_atom` uses the previously chosen subcomponent `0.attn.o:268` on the removal side, for German only. This reuses a previously chosen intervention with its selection fixed by the earlier experiment.

As in the main results, Table 5 gives each alternative edit's achieved target rise next to its contrasts, averaged over the same draws.

Table 5: Alternative subcomponent-selection comparisons. Target rise: the alternative edit's achieved target CE rise, averaged over the draws its contrast used, on confirmation / development data. The selected subcomponent's rise is in Table 1. Protected: VPD subcomponent minus reference, difference in protected CE rise, with simultaneous 95% intervals adjusted within this comparison family. These comparisons are descriptive. All values in nat/token. Draws used: the draws in which the reference has a matched, usable edit. With fewer than three the contrast is conditional, with none it has no estimate.

Pair	τ	Reference	Target rise	Protected, confirmation	Protected, development	Draws used
de → nl	0.5	OMP from full-matrix fine-tuning	0.49 / 0.51	−0.072 [−0.080, −0.065]	−0.071 [−0.080, −0.063]	0, 1, 2
de → nl	0.5	OMP from rank-one LoRA	0.48 / 0.50	−0.024 [−0.027, −0.022]	−0.024 [−0.027, −0.021]	0, 1, 2
de → nl	0.5	CI-difference selection	0.52 / 0.52	−0.776 [−0.815, −0.737]	−0.789 [−0.823, −0.756]	0, 1, 2
de → nl	0.5	Previously chosen subcomponent	0.48 / 0.49	0 (identical edit)	0 (identical edit)	0, 1, 2
de → nl	1	OMP from full-matrix fine-tuning	0.98 / 1.01	−0.134 [−0.145, −0.123]	−0.133 [−0.145, −0.120]	0, 1, 2
de → nl	1	OMP from rank-one LoRA	0.98 / 1.00	−0.045 [−0.048, −0.041]	−0.044 [−0.048, −0.040]	0, 1, 2
de → nl	1	CI-difference selection	0.98 / 0.99	−1.369 [−1.428, −1.311]	−1.414 [−1.464, −1.365]	0, 1, 2
de → nl	1	Previously chosen subcomponent	0.98 / 1.00	0 (identical edit)	0 (identical edit)	0, 1, 2
es → pt	0.5	OMP from full-matrix fine-tuning	0.51 / 0.51	−0.185 [−0.202, −0.169]	−0.182 [−0.197, −0.167]	0, 1, 2
es → pt	0.5	OMP from rank-one LoRA	0.51 / 0.51	−0.185 [−0.202, −0.169]	−0.182 [−0.197, −0.167]	2
es → pt	0.5	CI-difference selection	0.45 / 0.46	−0.484 [−0.508, −0.460]	−0.494 [−0.520, −0.469]	0, 1, 2

Pair	τ	Reference	Target rise	Protected, confirmation	Protected, development	Draws used
es → pt	1	OMP from full-matrix fine-tuning	no edit	no estimate	no estimate	none
es → pt	1	OMP from rank-one LoRA	no edit	no estimate	no estimate	none
es → pt	1	CI-difference selection	0.96 / 0.98	−1.046 [−1.085, −1.007]	−1.083 [−1.126, −1.039]	0, 1, 2
fr → it	0.5	OMP from full-matrix fine-tuning	0.50 / 0.52	−0.425 [−0.448, −0.403]	−0.436 [−0.457, −0.415]	0, 1, 2
fr → it	0.5	OMP from rank-one LoRA	0.47 / 0.49	−0.618 [−0.653, −0.582]	−0.629 [−0.662, −0.596]	0
fr → it	0.5	CI-difference selection	0.50 / 0.51	−0.483 [−0.511, −0.455]	−0.489 [−0.513, −0.464]	0, 1, 2
fr → it	1	OMP from full-matrix fine-tuning	1.01 / 1.05	−0.470 [−0.493, −0.447]	−0.479 [−0.498, −0.460]	0, 2
fr → it	1	OMP from rank-one LoRA	0.99 / 1.04	−1.236 [−1.307, −1.165]	−1.271 [−1.329, −1.213]	0
fr → it	1	CI-difference selection	0.95 / 1.00	−1.339 [−1.403, −1.276]	−1.367 [−1.422, −1.312]	0, 2

- **Identical edits.** For German, the previously chosen subcomponent is the selected subcomponent with the same sign and dose, so its contrast is exactly zero. `omp_lora` also picks `0.attn.o:268` in draws 0 and 1, where its edit equals the selected one. In draw 2 it picks `0.attn.v:271`, the subcomponent that `omp_finetune` picks in all German draws. Only draw 2 contributes a nonzero contrast. The reported estimate averages all three draws. For French, `omp_finetune` in draw 0 picks `0.attn.o:640`, the selected subcomponent.
- **Spanish.** `omp_finetune` picks `3.mlp.up:1435` in all draws, which needs a dose of 1.927 for $\tau = 0.5$ and does not reach $\tau = 1$ within the limit. `omp_lora` reaches $\tau = 0.5$ only in draw 2, where it picks the same subcomponent, so its estimate equals `omp_finetune`'s while its status is conditional. Neither has an edit at $\tau = 1$.
- **French.** The pursuit of the full-matrix fine-tuning update picks a different subcomponent in each draw. `omp_lora` has an edit only in draw 0: in draws 1 and 2 no checkpoint of the LoRA across all matrices met its checkpoint rule. In draw 1, `omp_finetune` and `ci_difference` do not reach $\tau = 1$.

D Reproducibility and availability

Code, saved measurements and reproduction instructions are on GitHub. Saved per-date measurements support analysis replay. The confirmatory token windows and texts are not released, so that model evaluation cannot be rerun from this checkout. The guide describes the available model procedures.